

Harshita Jogi

Software Engineer · Sunnyvale, CA · harshitajogi2001@gmail.com · linkedin.com/in/harshita-jogi-563227215 · github.com/HarshitaJogi

Open to full-time roles starting 2027. Graduating May 2027.

EDUCATION

Northeastern University, Boston, MA

Sep 2025 – May 2027

Master of Science in Computer Science · GPA 3.9/4.0

Coursework: Algorithms, Artificial Intelligence, Database Management, Object-Oriented Design, Distributed Systems

Awards: Academic Excellence Scholarship, International Impact Award

University of Mumbai, Mumbai, India

Aug 2020 – May 2024

Bachelor of Engineering in Electronics · GPA 9.04/10

EXPERIENCE

Nokia · Sunnyvale, CA

May 2026 – Present

Software Engineer Co-op, Test Automation & AI Tooling

- Extended a 50K+ line Python test framework, porting routing protocol test suites to a new router platform on live hardware.
- Built an LLM agent to automate test porting, grounded via internal-doc tool calls and a failure checklist mined from past debugging.
- Redesigned the agent into staged, dependency-ordered runs after live evaluation, adding false-pass checks and human review gates.

Network Science Institute, Northeastern University · Boston, MA

Jul 2025 – Jul 2026

Research Assistant

- Created a Python pipeline that uses GPT-4.1 to automate annotation of 1M+ biomedical research papers.
- Raised annotation accuracy from 60% to 98% via prompt iteration driven by false-positive and false-negative analysis against ground truth.
- Reduced GPT-4.1 token usage with structured output schemas and parsing logic, without sacrificing accuracy.
- Fine-tuned SciBERT for biomedical named entity recognition (NER), reaching 97% F1 on research-tool extraction.

MSCI, Morgan Stanley Capital International · Mumbai, India

Jan 2024 – Aug 2025

Technology Analyst

- Migrated 15+ APIs from Azure to GCP via Kubernetes and Docker, cutting latency by 23% and infrastructure costs by 40%.
- Engineered Spring Boot REST APIs for client-facing services, achieving 82% test coverage via JUnit and Gatling.
- Developed Databricks ETL pipelines using PySpark and Pandas, moving 5TB of data from OracleDB to BigQuery.
- Automated CI/CD release pipelines with SonarQube scanning, remediating security vulnerabilities before production.

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Research Intern

- Enhanced a YOLOv9 model with a modified architecture for better feature extraction on agricultural imagery.
- Reached 86% detection accuracy using multi-scale feature fusion and optimized loss functions.
- Applied model quantization for edge deployment on NVIDIA Jetson.

PROJECTS

Bitgig, Berkeley × DeepMind Hackathon · [live demo](#)

Sep 2026

- Architected a human-in-the-loop video data pipeline on Cloud Run and Cloud Storage, keeping datasets in-network with Gemini.
- Integrated the Gemini API to pre-segment lab and medical video into SOP steps, so experts correct drafts, not raw footage.
- Implemented consensus QC across expert reviews, routing disagreements to resolution before labels become training data.
- Shipped the expert-review interface in Next.js, React, and TypeScript on Vercel, from video and SOP upload to verified labels.

Autonomous Drone-Based Precision Agriculture System, IEEE AEES DSTEI funded research (\$25,000),

Research Lead

Jul 2023 – Dec 2023

- Built a TensorFlow pipeline using NDVI for region-of-interest detection, enabling accurate geospatial crop mapping.
- Deployed a quantized CNN on an NVIDIA Jetson-mounted drone, achieving real-time leaf classification at 15m altitude.
- Published 2 IEEE SPACE papers and filed a patent.

Event Scheduling & Management System, Java · Coursework

Nov 2025 – Dec 2025

- Built an MVC calendar engine with the Command pattern, supporting 18+ operations via regex-based parsing.
- Developed a multi-calendar system with timezone conversion, recurring events, and conflict detection.
- Designed a pluggable export module using Strategy and Factory patterns for CSV and iCal formats.

TryBud, Harvard Hack-o-Ween, Oct 2025. Blockchain job-verification platform. 2nd place, \$3,600 prize.PUBLICATIONS AND PATENT

K. Gharat, H. Jogi, K. Gode, K. Talele, S. Kulkarni, and M. H. Kolekar, “Enhanced Detection of Maize Leaf Blight in Dynamic Field Conditions Using Modified YOLOv9,” *2024 IEEE Space, Aerospace and Defence Conference (SPACE)*, pp. 140–143, 2024.

K. Gode, K. Gharat, H. Jogi, A. Sapkal, R. Thakar, S. Vishwakarma, K. Talele, and S. Kulkarni, “Multi-Stage UAV-Based System for Scalable and Accurate Crop Health Monitoring,” *2024 IEEE Space, Aerospace and Defence Conference (SPACE)*, pp. 652–655, 2024.

Patent filed: Automated Crop Health Monitoring System Using Unmanned Aerial Vehicle.

LLM systems: LLM agents, Model Context Protocol, Prompt engineering, LLM evaluation, GPT-4.1, Gemini API, Cursor

Machine learning: SciBERT, NER, TensorFlow, YOLOv9, Model quantization, NVIDIA Jetson, Prodigy

Data: Databricks, PySpark, Pandas, BigQuery, OracleDB, ETL, Data modeling

Backend: Spring Boot, REST APIs, JUnit, Gatling

Cloud & DevOps: GCP, Cloud Run, Cloud Storage, Azure, Docker, Kubernetes, CI/CD, SonarQube, Vercel

Languages & web: Python, Java, TypeScript, React, Next.js

Also familiar with: C++, SQL, JavaScript, PostgreSQL, AWS, Git

ALSO

Kovida degree in Bharatanatyam, Nalanda Dance Research Center. Trinity College London, Communication Skills Grade 5 with Distinction.